ANSWER SCHEME

PART A

1. (a)

$$\lim_{x\to\infty}\frac{5x^3+4x-9}{7-3x^3}=\lim_{x\to\infty}\frac{\frac{5x^3}{x^3}+\frac{4x}{x^3}-\frac{9}{x^3}}{\frac{7}{x^3}-\frac{3x^3}{x^3}}=\lim_{x\to\infty}\frac{5+\frac{4}{x^2}-\frac{9}{x^3}}{\frac{7}{x^3}-3}=-\frac{5}{3}$$

(b) For $\lim\limits_{x\to 2} g(x)$ to exists

$$\lim_{x\to 2^-} g(x)=\lim_{x\to 2^+} g(x)$$

$$\lim_{x\to 2^-} m^2-8=\lim_{x\to 2^+}\frac{x-2}{\sqrt{2x}-2}$$

$$m^2-8=\lim_{x\to 2^+}\frac{x-2}{\sqrt{2x}-2}\times\frac{\sqrt{2x}+2}{\sqrt{2x}+2}$$

$$m^2-8=\lim_{x\to 2^+}\frac{(x-2)\left(\sqrt{2x}+2\right)}{2x-4}$$

$$m^2-8=\lim_{x\to 2^+}\frac{\sqrt{2x}+2}{2}$$

$$m^2-8=2$$

$$m^2=10$$

$$m=\pm\sqrt{10}$$

2. (a)

$$f'(x)=\lim_{h\to 0}\left(\frac{1}{x+h+3}-\frac{1}{x+3}\right)\times\frac{1}{h}$$

$$f'(x)=\lim_{h\to 0}\left(\frac{x+3-x-h-3}{(x+h+3)(x+3)}\right)\times\frac{1}{h}$$

$$f'(x)=\lim_{h\to 0}\frac{-h}{(x+h+3)(x+3)}\times\frac{1}{h}$$

$$f'(x)=\lim_{h\to 0}-\frac{1}{(x+h+3)(x+3)}$$

$$f'(x)=-\frac{1}{(x+3)^2}$$

(b)

$$y=x^2e^{-3x}$$

$$\frac{dy}{dx}=2xe^{-3x}-3x^2e^{-3x}=e^{-3x}\left(2x-3x^2\right)$$

$$\frac{d^2y}{dx^2}=-3e^{-3x}\left(2x-3x^2\right)+e^{-3x}\left(2-6x\right)=e^{-3x}\left(9x^2-12x+2\right)$$

3. $$y = x^2 - \frac{6}{x}$$

$$\frac{dy}{dx} = 2x + \frac{6}{x^2}$$

At the stationary point,

$$\frac{dy}{dx} = 0 \quad \rightarrow \quad 2x + \frac{6}{x^2} = 0$$

$$2x^3 = -6$$

$$x = (-3)^{\frac{1}{3}} = -1.442$$

$$y = (-1.442)^2 - \frac{6}{-1.442} = 6.240$$

Stationary points is (-1.442, 6.240)

$$\frac{d^2y}{dx^2} = 2 - 12x^{-3}$$

$$\frac{d^2y}{dx^2}\Big|_{x=-1.442} = 2 - \frac{12}{(-1.442)^3} = 6.002 \quad (> 0)$$

$\therefore$ $(-1.442, 6.240)$ is the minimum point.

PART B

1. (a)

$$z_1^2 = (-i)^2 = -1 @ z_1^2 = -1$$
$$\overline{z_2} = 2 - i\sqrt{3}$$

(b)

$$W = \frac{1 - i\sqrt{3}}{-i} \times \frac{i}{i} = \frac{i - i^2\sqrt{3}}{1} = \sqrt{3} + i$$
$$|W| = \sqrt{\left(\sqrt{3}\right)^2 + (1)^2} = 2$$
$$\arg(W) = \tan^{-1}\left|\frac{1}{\sqrt{3}}\right| = \frac{\pi}{6} @ 0.524$$

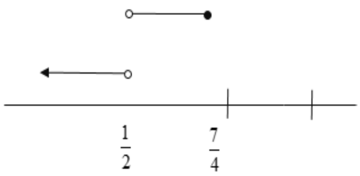
2. (a)

$$\log_2 2x = \frac{2\log_2(x+4)}{\log_2 4}$$
$$\log_2 2x = \frac{2\log_2(x+4)}{2}$$
$$\log_2 2x = \log_2(x+4)$$
$$2x = x + 4$$
$$x = 4$$

(b) $\dfrac{x-3}{2x-1} \geq \dfrac{1}{2} \qquad \cup \qquad \dfrac{x-3}{2x-1} \leq -\dfrac{1}{2}$

$\dfrac{-5}{2(2x-1)} \geq 0 \qquad\qquad \dfrac{4x-7}{2(2x-1)} \leq 0$

$x < \dfrac{1}{2} \qquad\qquad \dfrac{1}{2} < x \leq \dfrac{7}{4}$ (By sign table)



$$\left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{7}{4}\right]$$

3. (a) $f(x) = \dfrac{1}{1+\dfrac{1}{1+\dfrac{1}{x}}} = \dfrac{1}{1+\dfrac{x}{x+1}} = \dfrac{x+1}{2x+1}$

$$f\left(\frac{1}{2}\right) = \frac{\frac{1}{2}+1}{2\left(\frac{1}{2}\right)+1} = \frac{3}{4}$$

(b) $\dfrac{1}{1+\dfrac{1}{1+\dfrac{1}{x}}} \qquad \Rightarrow \dfrac{1}{x},\ x \neq 0$

$\dfrac{1}{1+\dfrac{x}{x+1}} \qquad \Rightarrow \dfrac{x}{x+1}, x \neq -1$

$\dfrac{x+1}{2x+1} \qquad \Rightarrow \dfrac{x+1}{2x+1}, x \neq -\dfrac{1}{2}$

$\therefore$ the numbers are $0, -1, -\dfrac{1}{2}$

4. (a) $D_f = \left(-\dfrac{1}{2}, \infty\right)$

$R_f = (-\infty, \infty)$

(b) $f\left[f^{-1}(x)\right]=x$

$\ln\left[2f^{-1}(x)+1\right]=x$

$2f^{-1}(x)+1=e^x$

$f^{-1}(x)=\dfrac{e^x-1}{2}$

$D_{f^{-1}}=(-\infty,\infty)$

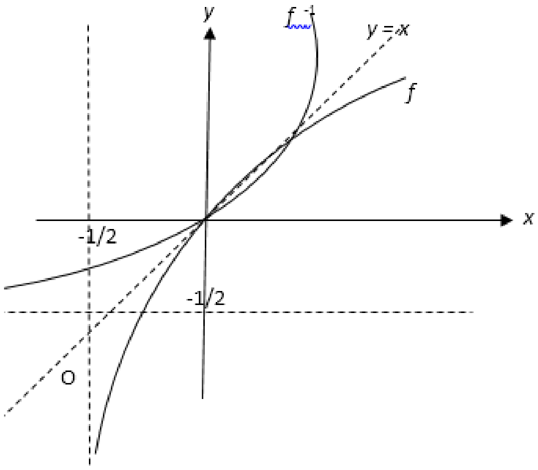
$R_{f^{-1}}=\left(-\dfrac{1}{2},\infty\right)$

$f^{-1}(x)=0 \;\Rightarrow\; \dfrac{e^x-1}{2}=0$

$e^x=1$

$x=0$

(c)



5. (a) For $g(x)$ is discontinuous at $x=8$

$g(8)\neq \lim\limits_{x\to 8^+} f(x)$

$3\neq \lim\limits_{x\to 8^+} \dfrac{|8-x|}{x-8}+k$

$3\neq \lim\limits_{x\to 8^+} \dfrac{-(8-x)}{x-8}+k$

$3\neq \lim\limits_{x\to 8^+} 1+k$

$3\neq 1+k$

$k\neq 2$

(b) (i) $f(3)=-6$

(ii) $\lim\limits_{x\to 3^+}\dfrac{162x-54x^2}{x^3-27}=\lim\limits_{x\to 3^+}\dfrac{54x(3-x)}{(x-3)(x^2+3x+9)}=\lim\limits_{x\to 3^+}-\dfrac{54x}{(x^2+3x+9)}=-6$

$\lim\limits_{x\to 3^-}\dfrac{9-x^2}{x-3}=\lim\limits_{x\to 3^-}\dfrac{(3-x)(3+x)}{x-3}=\lim\limits_{x\to 3^-}-\dfrac{(3+x)}{x-3}=-6$

$\lim\limits_{x\to 3} f(x)=-6$

(iii) $f(3)=\lim\limits_{x\to 3} f(x)=-6$

$\therefore f(x)$ is continuous at $x=3$.

6. (a) $x^2y=\sin\left(2x^2+\pi\right)$

$$2xy+x^2\frac{dy}{dx}=4x\cos\left(2x^2+\pi\right)$$

$$2y+2x\frac{dy}{dx}+2x\frac{dy}{dx}+x^2\frac{d^2y}{dx^2}=4\cos\left(2x^2+\pi\right)-16x^2\sin\left(2x^2+\pi\right)$$

$$2y+4x\frac{dy}{dx}+x^2\frac{d^2y}{dx^2}=4\cos\left(2x^2+\pi\right)-16x^2\left(x^2y\right)$$

$$2y+16x^4y+4x\frac{dy}{dx}+x^2\frac{d^2y}{dx^2}=4\cos\left(2x^2+\pi\right)$$

$$2y\left(1+8x^4\right)+4x\frac{dy}{dx}+x^2\frac{d^2y}{dx^2}=4\cos\left(2x^2+\pi\right)$$ Shown

(b) (i)

$x=2t-t^{-1}$

$\dfrac{dx}{dt}=2+\dfrac{1}{t^2}$

$=\dfrac{2t^2+1}{t^2}$

$y=2t+t^{-1}$

$\dfrac{dy}{dt}=2-\dfrac{1}{t^2}$

$=\dfrac{2t^2-1}{t^2}$

$$\frac{dy}{dx}=\frac{2t^2-1}{t^2}\times\frac{t^2}{2t^2+1}$$

$$\frac{dy}{dx}=\frac{2t^2-1}{2t^2+1}$$

(ii)

$$\frac{d^2y}{dx^2}=\frac{(2t^2+1)(4t)-(2t^2-1)(4t)}{(2t^2+1)^2}\times\frac{t^2}{(2t^2+1)}$$

$$\frac{d^2y}{dx^2}=\frac{8t^3}{(2t^2+1)^3}=\frac{8}{y^3}$$ shown

$y=2t+\dfrac{1}{t}=\dfrac{2t^2+1}{t}$

$y^3=\left(\dfrac{2t^2+1}{t}\right)^3$

$\dfrac{1}{y^3}=\left(\dfrac{t}{2t^2+1}\right)^3$

7. (a) Volume of the cylinder,

$$V = \pi r^2(2h)$$
$$= 2\pi(a^2 - h^2)h$$

hence,

$$V = 2\pi a^2 h - 2\pi h^3$$
$$\frac{dV}{dh} = 2\pi a^2 - 6\pi h^2$$
$$\frac{d^2V}{dh^2} = -12\pi h$$

(b)

$$\frac{dV}{dh} = 0$$
$$2\pi a^2 - 6\pi h^2 = 0$$
$$h = \frac{a}{\sqrt{3}}$$
$$when\ h = \frac{a}{\sqrt{3}},\ \frac{d^2V}{dh^2} = -12\pi\left(\frac{a}{\sqrt{3}}\right) < 0, because\ a > 0$$

$$V_{\max} = 2\pi\left(a^2 - \left(\frac{a}{\sqrt{3}}\right)^2\right)\left(\frac{a}{\sqrt{3}}\right)$$
$$= 2\pi\left(\frac{2a^2}{3}\right)\left(\frac{a}{\sqrt{3}}\right)$$
$$= \frac{4\pi a^3}{3\sqrt{3}}$$
$$= \frac{4\sqrt{3}\pi a^3}{9}$$